AIM

MD.090 DEVELOPMENT ENVIRONMENT

<Company Long Name>

<Subject>

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Contents

Introduction

Purpose

The purpose of the **Development Environments** document is to describe the application installations, application server and database server instances needed to support the development and testing of custom extensions. It includes:

- Definition of the tiering architecture that will be used.
- Standards used for describing the physical architecture.
- Strategy for designing the physical database including major operational factors, use of striping and disk arrays (RAID), efficient utilization of disk space and facilitation of database administration tasks.
- Specification of Key Server Parameters.
- Mapping of the logical database objects to tablespaces and physical datafiles.
- Estimation of the I/O throughput requirements.

Suggestion: Specific character set and the availability globalizations or localizations may affect the architecture requiring more than one database or applications server installation.

Environment - Development

Update this section to reflect the nature of the facilities management arrangement. For example, the corporate data center may be hosted or facility managed by an outside party.

Environment Hosting

The development environment is owned, managed and controlled by <Company Long Name>. This data center is located at: <Data Center Location>

Sharing of Environments

The extension development activities will share the same environment used by:

- Data Conversion Development and Testing
- Business Process Review/Development, Business Requirements Definition
- Business Requirements Mapping
- Project Team Learning (Training)
- User Learning (Training)
- Business Systems Testing
- Performance Testing
- Application and Technical Architecture Design/Development

Seeding of Configuration and Sample Data

The development environment used to support development of application extensions will be seeded periodically with what will be the production setups (configuration) of the applications. From time to time these setups will be updated to reflect what is planned for the final production environment.

When setups are re-synchronized with the production environment, an assessment will be undertaken to identify any application extensions that have completed the testing cycle, that will require re-testing. This approach allows for parallel development of the application extensions and production setups. However, careful change management will be required to identify these occurrences.

Database Tier

The database tier holds all data and data-intensive programs, and processes all SQL requests for data. The database tier includes the Oracle Database Server, the administration server, and the concurrent processing server. The concurrent processing server executes traditional batch processing tasks such as transactional interfaces and reports.

By definition, platforms in this tier do not communicate directly with applications users, but rather with machines on the application tier that mediate these communications, or with other servers on the database tier.

Database Servers

The database server contains the transactional and schema data associated with the Oracle applications. The platform that supports this server contains two sets of objects. One set is the necessary files that support a running database server, and the other set is a series of database files that contain the actual database data.

Suggestion: One platform may support more than one database instance such as PRODuction and TEST. One set of database files (executables) may support more than one database instances. There may some restrictions depending on the operating and Oracle Database Server releases that you are running.

The following Oracle Database Server instances will be established to support the application extension development activities of the project:

Oracle Database Instance	Platform / DNS	Platform O/S	Server Release	Function
DEV	Compaq Proliant 640 (trn1)	Windows 3000 Server	Oracle 9.5*i*	Application Development Environment

Naming Standards

This section describes the naming standards that will be used for objects and files relevant to the database architecture:

- File System Mount Points
- Oracle File Categories
- Database Files
- Tablespaces

<u>File System Mount Points</u>

The standard for File System Mount Points is:

```
Standard: '/u[01-99]/'
Example: '/u03/'
```

File Categories

The standard for File Categories is:

```
Standard: ' ''
Example: ' '
```

Database Files

The standard for Database File Names is:

```
Standard: ''
Example: ''
```

Tablespaces

The standard for Tablespace Names is:

```
Standard: ''
Example: ''
```

Mount PointsThe standard for File System Mount Points is:

```
Standard: ''
Example: ''
```

Oracle Database Instance - <ORACLE_SID>

Key Init.ora Listing

The following table documents the values of other key database init.ora parameters:

Parameter Name	Parameter Value	Description
_undo_optimizer_changes	TRUE	

Additional Database Server Configuration Details

Below are additional configuration details for the Database Server:

Parameter	Value
What is the platform DNS name	CONV
What Type of Hardware Platform and OS is being used	Sun Solaris 9.5
Oracle Block Size	
Net8 Listener Port	1521
Net8 Report Procedure Call (RPC) Server Port	1522
Character Set used for database creation	WE8ISO8859P1
Location of Oracle Home Directory	/u01/app/oracle/product/905
Location of Oracle Base Directory	/u01/app/oracle
Is the Optimal Flexible Architecture Standard Being Used	Yes

Tablespace Partitioning

This section describes the basic partitioning of the tablespaces within the production database instances. The partitioning is designed to satisfy <Company Short Name>'s database architecture strategy.

Segment Type	Tablespace	Comments
System		

Segment Type	Tablespace	Comments
Temporary		
Rollback		
Tools		
Misc User		
Data		
Index		

Tablespace Storage Parameters

The following table shows the tablespaces for this Oracle Applications database instance:

		Storage Parameters					Number of	Filesystems				
Tablespace	Size (Mb)	init	nxt	pct	min	max	Datafiles	/u1	/u2	/u3	/u4	/u5
Redo Logs												
SYSTEM												
RBS												
TEMP												
USERS												
Totals	0						0	0	0	0	0	0

Tablespace File Mapping

This section describes the mapping of the tablespace structures onto physical storage files.

Tablespace	Size (MB)	Filesystem	Directory	Datafile
RSEGS	105	sd7h	/ora2/ORACLE/proddb	prod_rseg1.dbf
AOLTAB	105	sd4g	/ora3/ORACLE/proddb	prod_aoltab1.dbf
ARTAB	210	sd4g	/ora3/ORACLE/proddb	prod_artab1.dbf
GLTAB	262	sd4g	/ora3/ORACLE/proddb	prod_gltab1.dbf
POIND	157	sd4g	/ora3/ORACLE/proddb	prod_poind1.dbf
SHAREIND	52	sd4g	/ora3/ORACLE/proddb	prod_sharedind1.dbf
AOLIND	52	sd5g	/ora4/ORACLE/proddb	prod_aolind1.dbf

Administrative Servers

The administration server is the machine from which you maintain the data in your Applications database. There are three types of operations you will carry out here, each using a different program:

Installing and Upgrading the Database

This process is only performed when installing a new release, or upgrading to a new minor or major release.

Applying Applications Database Updates

Most maintenance patches will consist of new files and scripts that update database objects.

Maintaining the Applications Data

Some features, such as MultiLingual Support and Multiple Reporting Currencies, require regular maintenance to ensure updates are propagated to the additional schemas used by these features

Suggestion: The administration server is the most infrequently used, compared to other servers in the Applications multi-tier environment, and has the smallest computing requirements. You therefore should not need to have more than one administration server for your installation.

The following Administrative Servers will be established to support the application extension development activities of the project:

Oracle Database Instance	Platform / DNS	Platform O/S	Server Release	Function
DEV	Sun E-950 (prod1)	Solaris 5.2	Oracle 9.5*i*	Applications Development Environment

Concurrent Processing Servers

Most interaction with applications data occur through Applications forms. There are also reporting programs, however, that periodically need to be run. These programs may contain a very large number of computations, so to ensure that they do not interfere with interactive operations, they can be configured to run on a separate machine called the concurrent processing server.

Processes that run on the concurrent processing server are called concurrent programs, and operate in the background while users continue to work on other tasks. These programs are typically executables written in C or reports written using Oracle Reports.

A request to run a concurrent program is submitted through applications forms, which insert the request into a database table. When the table is read by a monitoring process, the request is assigned to one of several concurrent managers (also referred to as workers) running on the concurrent processing server. The concurrent manager processes the request (which may involve calling another program, for example, or running Oracle Reports), and generates log and output files, which are stored on the concurrent processing server.

Additional Information: Overview of Concurrent Processing, Oracle Applications Systems Administrator's Guide

Suggestion: More than one server (one per platform) may be configured to provide parallel concurrent processing. Parallel concurrent processing allows you to run concurrent managers on multiple servers in your database tier. This distributes the processing load and provides fault tolerance in case one or more servers fail.

The following Concurrent Processing Servers will be established to support the application extension development activities of the project:

Oracle Database Instance	Platform / DNS	Platform O/S	Server Release	Function
DEV	Sun E-950 (prod1)	Solaris 5.2	Oracle 9.5*i*	Applications Development Environment

<Platform Name>

Platform Configuration

Describe in detail the configuration of this platform.

Disk Device Map

The worksheet below documents the disk device map on the hardware platform designated as <Platform Name>.

Device	Filesystem	Available Space	
C0S0S0	/u1	4,354	Mb
C0S0S1	/u2	654	Mb
			Mb
			Mb
			Mb
			Mb
			Mb
			Mb
			Mb

I/O Subsystem Configuration

Describe the configuration of the I/O subsystem for this platform.

-
-
-

Application Tier

The application servers form the middle tier between the desktop clients and database servers. They provide load balancing, business logic, and other functionality.

Attention: In installations that use multiple application servers, only one needs to run the Oracle Web Application Server software.

Forms Servers

The forms server is a specific type of application server that hosts the Oracle Forms Server engine. The Oracle Forms Server is an Oracle Developer component that mediates between the desktop client and the Oracle Database Server, displaying client screens and causing changes in the database records based on user actions. Data is cached on the forms server and provided to the client as needed, such as when scrolling through multiple order lines. The forms server exchanges messages with the desktop client across a standard TCP/IP network connection.

Suggestion: Load Balancing Among Forms Servers provides automatic load balancing among multiple application servers. In a load-balancing configuration, a single coordinator called the Metrics Server is on one application server. Metrics Clients located on the other application servers periodically send load information to the Metrics Server so it can determine which has the lightest load. When a client issues a request to download the Forms client applet, the Metrics Server provides the name of the least-loaded host for the applet to connect to.

The following Oracle Web Application (Forms) Server instances will be established to support the application extension development activities of the project:

Platform / DNS	Platform O/S	Server Release	Metric	Forms Port	Forms Server Port	Function
Sun E-950 (was1)	Solaris 5.2	2.5	Server	8000	9000	Applications Server, Load Management Machine (Production)
Sun E-950 (was2)	Solaris 5.2	2.5	Client	8000	9000	Applications Server (Production)
Sun E-950 (was3)	Solaris 5.2	2.5	Client	8000	9000	Applications Server (Production)

Attention: Automatic failover capabilities are inherent in this load-balancing system. If an application server becomes unavailable for any reason, the Metrics Server ceases to route requests to the server until it comes back online. While the application server is offline, requests are routed to one of the other application servers.

Oracle Applications File System

The Oracle Applications File system contains all of the forms and executables that execute of the applications tier of the architecture.

Owner of the File System

The owner of the Oracle Applications File System is: <Userid> (Example: applmgr)

Additional Forms Server Configuration Details

Below are additional configuration details for the Forms Server:

New Cartridge Configuration Settings

Parameter	Value
Cartridge Name:	<Oracle SID>
Object Path:	$ORACLE_HOME/lib/f65webc.so
Entry Point:	form_entry

Virtual Paths

Parameter	Value
/Oracle SID	$ORACLE_HOME/lib
Name of DAD to be used:	<Oracle SID>
Protect PL/SQL Agent:	TRUE
Authorized Ports:	<Web Listener Port>

Cartridge Configuration Screen Values

Parameter	Value
baseHTML	$OA_HTML/afsampled.htm
serverPort	<Forms Port>
userid	APPS/APPS@${ORACLE_SID}
fndnam	APPS

Web Servers

The web server is another type of application server, which runs an HTTP listener. The HTTP listener (also called a web listener) is a component of an HTTP server, such as Microsoft Internet Information Server, or Netscape Enterprise Server. This listener accepts incoming HTTP requests (or URLs) from desktop clients, via the web browser or appletviewer. These requests are either immediately processed—for example, by returning an HTML document—or are passed on to the Oracle Web Application Server, which also resides on the same platform.

The following Web Servers will be established to support the application extension development activities of the project:

Server Name	Platform / DNS	Platform O/S	Server Release	UDP Port	Admin Port	Function
DEV	Sun E-950 (prod1)	Solaris 5.2	5.4	2649	8888	Applications Development Environment

Additional Web Server Configuration Details

Below are additional configuration details for the Web Server:

Virtual Directories

File System	Directory Flag	Virtual Path
$APPLTMP	NR	/OA_TEMP/
$APPL_TOP/doc/doc/	NR	/OA_DOC/
$APPL_TOP/html/html/	NR	/OA_HTML/
$APPL_TOP/html/html/bin/	CN	/OA_HTML/bin/
$ORACLE_HOME/forms65/java/	NR	/OA_JAVA/
$ORACLE_HOME/forms65/java/oracle/apps/media/	NR	/OA_MEDIA/

Java Cartridge Path

Virtual Path	Application	Physical Path
/OA_JAVA_SERV	JAVA	$ORACLE_HOME/forms65/java

DAD Settings

Parameter	Value
DAD Name:	<Oracle SID>
Database User:	APPS
Identified by:	APPS
Database User Password:	APPS

Parameter	Value
ORACLE HOME:	$ORACLE_HOME
Net8 Service:	<Oracle SID>
NLS Language:	American

PL/SQL Cartridge Settings

Parameter	Value
Name of PL/SQL Agent:	<Oracle SID>
Name of DAD to be used:	<Oracle SID>
Protect PL/SQL Agent:	TRUE
Authorized Ports:	<Web Listener Port>

PL/SQL Virtual Path

Virtual Path	Application	Physical Path
/<Oracle SID>/plsql	PLSQL	$ORACLE_HOME/ows/6.0/bin

<Platform Name>

Platform Configuration

Describe in detail the configuration of this platform.

Disk Device Map

The worksheet below documents the disk device map on the hardware platform designated as <Platform Name>.

Device	Filesystem	Available Space	
C0S0S0	/u1	4,354	Mb
C0S0S1	/u2	654	Mb
			Mb
			Mb
			Mb
			Mb
			Mb
			Mb
			Mb

I/O Subsystem Configuration

Describe the configuration of the I/O subsystem for this platform.

-
-
-

Desktop Client Tier

The desktop client runs a Java applet using a Java-enabled web browser or appletviewer. The applet sends user requests to the forms server and handles such responses as screen updates, pop-up lists, graphical widgets, and cursor movement. It can display any Oracle Applications screen and supports field-level validation, multiple coordinated windows, and data entry aids like lists of values. A web browser or appletviewer manages the downloading and storage of the Forms client applet on each user's desktop. They also supply the Java Virtual Machine (JVM) that runs the Forms client applet.

Security: Security features ensure that the forms server only accepts connections from "certified" Forms clients bearing the Oracle Applications signature. For additional security, all communication between the Forms client applet and forms server is encrypted using the RSA RC4 40-bit standard form of encryption.

Web Browsers

Below are the web browsers that will be supported by the project during deployment:

- <list Browsers: Name, OS, Version>
-
-

Web Appliances

Web appliances are devices that are capable of supporting the presentation and execution of a Java applet. These may be single purpose devices or mobile devices. Below are the web appliances that will be supported by the project during deployment:

<list web appliances: Name, Manufacturer, Model Number>

Applet Viewers

Applet views are run time environments that run on an operating system and present and execute Java applets. Below are the applet viewers that will be supported by the project during deployment:

<list applet viewer: Name, OS, Version>

Other Applications

The other applications that will be required to support the application extensions development activities of this project:

<List other applications>

Automated Development Tools

Below are the access, security, and additional tablespace requirements for each automated development tool you are using to facilitate the development process.

Development Tool: <Development Tool>

Access Requirements:

-
-
-

Security Requirements:

-
-
-

Additional Tablespace Requirements:

-
-
-

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date